HARVARD T.H. CHAN | SCHOOL OF PUBLIC HEALTH

BST 219: Core Principles of Data Science
Fall 2026
T/Th 9:45-11:15am, Kresge G2 (except 12/1 & 12/3 in FXB G12)

Instructor Information

Faculty

Dr. Heather Mattie
Lecturer on Biostatistics
Co-Director of the Health Data Science Master's Program
Director of Community Engagement and Development
Director of the Research Education Core, The Partnership
Department: Biostatistics, HSPH
Office: Building 1, 4th Floor, Room 421A
Email: hemattie@hsph.harvard.edu
Office Hours: Thursdays 12-1pm (in person and on Zoom)
Note: email is **by far** the best way to contact me. I rarely check Canvas messages.

Teaching Fellows

Stephanie Armbruster – sarmbruster@g.harvard.edu
Monica Robles Fontan – mroblesfontan@g.harvard.edu
Kimberly Greco – kimberly.greco@g.harvard.edu

Office Hours

Day	Teaching Staff	Time	Location
Thursday	Heather	12-1pm	Building 1, 4th floor, room 421A and Zoom
TBD	Stephanie	TBD	TBD
TBD	Monica	TBD	TBD
TBD	Kimberly	TBD	TBD

Lab: TBD, Zoom

Credits

5 credits

Course Purpose and Description

Modern technology has led to the generation of unprecedented amounts of data, prompting the need to train researchers to leverage data for decision-making in public health and medicine. This course assumes no prior knowledge and serves as a gentle, practical introduction to data wrangling, visualizing, and modeling data using the R statistical programming language. We also emphasize the importance of reproducible research, effective data science communication, and the risk of algorithmic bias.

Pre-Requisites

None

Course Learning Objectives

Upon successful completion of this course, you should be able to:

- Write reproducible code using the statistical programming language R
- Clean and wrangle data for downstream analysis
- Perform exploratory data analysis, including data visualizations
- Apply machine learning models for regression and classification
- Interpret and communicate key data science project results

Course Readings

Required: students are encouraged to read the lecture documents and other resources available on the course Canvas site and the course GitHub repository.

Suggested:

1. R for Data Science (2nd edition, 2023; open access)
2. Storytelling with Data (available via Harvard Hollis)
3. ggplot2: Elegant Graphics for Data Analysis (3rd edition; open access)
4. Happy Git and GitHub for the useR (mainly chapter 12; open access)
5. An Introduction to Statistical Learning with R (aka ISR; free download)
6. R for Health Data Science (open access)

Course Structure

This course will be held synchronously and in-person. We encourage students to attend class and participate, but it is not required, and all lectures and lab sessions will be recorded and available on the course Canvas site.

The final grade for this course will be based on:

- 5 Homework Assignments (60%)
- 1 Take-home Midterm (15%)
- 1 Group project (25%)

Participation

Attendance and participation are not graded components of this course but are highly encouraged in order to get as much as possible out of this course.

Homework Assignments (60%)

All homework assignments will involve **writing code and communicating results**. Students must **submit the RMarkdown file (.Rmd file) and knitted html file (.html file)** associated with each assignment in their individual GitHub repository. A private repository for each assignment will be created for each student and will only be visible to the student and course teaching staff.

Each student is given two late days per homework assignment. A late day extends the individual homework deadline by 24 hours without penalty. No more than two late days may be used on any one homework assignment. Late days are intended to give students flexibility: students can use them for any reason, no questions asked. Students do not get any bonus points for not using late days. Homework submitted more than 48 hours after the deadline will incur a 10% (10 points) reduction for each additional day it is late. Students can only use late days for the individual homework deadlines; all other deadlines (e.g., project milestones, midterm exam) are firm.

The TFs must be able to knit submitted RMarkdown files. The penalty for not being able to knit a file while grading increases for each subsequent homework – see breakdown below. To avoid this, students should be sure to include relative paths to files, images, etc., rather than absolute paths (paths specific to their computer). Examples of how to include paths will be given in lecture and lab sessions. Students may also double check with the teaching staff before submitting assignments.

- 0 points for HW1
- 5 points for HW2
- 10 points for HW3
- 15 points for HW4
- 20 points for HW5

Students may ask questions about the assignments during lecture, but we ask that any questions about grading be directed to Dr. Mattie via email.

Students may work together on assignments but all responses to text questions must be in their own words and not copied from another student's assignment or from a generative AI tool like ChatGPT (see generative AI policy below).

In this course, we recognize the growing presence and impact of generative Artificial Intelligence (AI) tools (e.g., ChatGPT, Claude, Gemini, etc.) in academic and professional environments. These tools can be powerful aids in the learning process when used responsibly. The following guidelines are designed to ensure that generative AI is used ethically and effectively in your course assignments:

Permissible Use

1. Assistance: You may use generative AI tools to assist in brainstorming ideas, drafting outlines, generating code snippets, or seeking clarification on complex topics.
2. Supplemental Learning: These tools can serve as supplemental resources to explain concepts and provide additional examples that may aid in your understanding.

Impermissible Use

1. Plagiarism: You must not directly copy and submit AI-generated content as your own work. This includes text, code, images, or any other type of content that has not been adequately personalized or appropriately cited.
2. Substitution for Learning: Relying on AI tools to complete assignments in lieu of engaging with the course material is discouraged. Your primary goal should be to develop a deep understanding of the subject matter.

Transparency and Citation

1. Citation: Cite AI tools appropriately in your bibliography or reference sections in accordance with the citation style prescribed for the course.

Academic Integrity

1. Originality: Ensure that your submissions reflect your own understanding and synthesis of the material. AI tools should not replace your critical thinking and analytic skills.
2. Integrity: Any use of AI tools should uphold the principles of academic integrity outlined by Harvard TH Chan's academic policies.

By adhering to these guidelines, you will be able to harness the benefits of generative AI while maintaining the highest standards of academic integrity and personal learning. If you have any questions regarding the appropriate use of these tools, please feel free to reach out to Dr. Mattie for clarification.

Take-home Midterm (15%)

A take-home midterm will be distributed in the form of an RMarkdown file November 6th – 15th to test comprehension of course material. The exam will consist of multiple-choice questions that may or may not require writing code, coding questions, and short answer questions. All code used and text answers must be submitted using the RMarkdown file. Students must submit the exam via their individual GitHub repository by 11:59pm on November 15th. Students are encouraged to use lecture slides and code, lab material, homework assignments, and the Internet to work on the exam but may not work or consult with other students. The teaching staff will be available to answer any questions concerning the exam. **Students may not use any form of generative AI on the exam.**

Due to the unpredictable nature of COVID-19 students in need of extra time to complete the midterm should reach out to Student Affairs at StudentAffairs@hsph.harvard.edu. A staff member will work with you and Dr. Mattie to accommodate you. You can also contact Student Affairs if you have a learning disability that requires accommodations. We will ensure you are accommodated as needed.

Group Project (25%)

Students will work in small groups (4-5 students) on a data science project. The goal of the project is to go through the complete data science process to answer an assigned prompt. Groups will be given a dataset and series of questions to answer. Groups will design visualizations, provide summary statistics, build traditional statistical models and machine learning models, and communicate results. A full description will be made available in October. The final deliverable is an RMarkdown file and associated knitted html file uploaded to an assigned group GitHub repository.

Technical Assistance

Canvas

If the issue is Canvas-related (e.g., you can't figure out how to use something or a feature seems broken), first try the documentation located under the Help menu found on the left-hand side of each Canvas page. If the issue is not covered there, contact Instructure directly, also via the Help menu. You can e-mail, text, or speak live with them at any time day or night. If you cannot access Canvas to view the Help menu, you can reach Instructure by phone at +1 (844) 326-4466.

Harvard-Specific Issues
If the issue seems Harvard-specific (e.g., HUID or Harvard Key username authentication, email not working, etc.), contact the Helpdesk by emailing helpdesk@hsph.harvard.edu or calling +1 (617) 432-HELP (4357).

Harvard Chan Policies and Expectations
For policies related to Diversity and Inclusion, Academic Support, and Exam Policies, please review the Education Policies page.

Course Evaluation
The course evaluation system opens during the last week of term and stays open for three weeks. Evaluations may be accessed via Canvas or by following the links in invitation emails. Responses are confidential, and identifying information is never shared with us. **You will have early access to your grades once you have completed all available course evaluations**; we will receive our evaluation report after the evaluation window closes and only if all grades are submitted in my.harvard. More information on course evaluations is available on the Office of Educational Programs intranet page.

Your feedback makes a difference. We appreciate knowing what worked well and what needs improvement in our course. In addition, your feedback enables the Committee on Educational Policy to recognize outstanding teaching, to offer faculty development coaching and training opportunities, to aid department chairs in assigning teaching responsibilities, and to inform the deliberation of faculty appointment committees.

Community Norm of Non-Attribution
We strive to foster an environment at HSPH grounded in trust, mutual respect, diversity of viewpoints, free expression, open inquiry, and constructive dialogue. To support the highest levels of learning and critical thinking in our classrooms, everyone must feel comfortable sharing their perspectives, ideas, and lived experiences via respectful discussion of complex, sensitive and consequential public health questions and topics.

Our classroom will incorporate and adopt a community norm of Non-Attribution, modeled after the Chatham House Rule, permitting participants to mention the information discussed in class outside of class but prohibiting attribution of comments or ideas to a specific student, unless that person gives written or verbal permission.

Course Schedule

Lecture	Date	Topics	Assignments
1	1-Sep	Introduction to course, R, RStudio, RMarkdown	HW1 Assigned Due Friday, September 18th
2	3-Sep	Introduction to Git, GitHub and homework submission	
3	8-Sep	Basic R: data types, vectors, sorting, indexing	
4	10-Sep	Basic R: basic plots and programming basics	
5	15-Sep	Data wrangling: **`dplyr`** function, importing data, reshaping data	HW2 Assigned Due Friday, October 2nd
6	17-Sep	Data wrangling: combining data frames, parsing dates and times	
7	22-Sep	Data visualization: Introduction to **`ggplot2`**, fine-tuning visualizations	HW3 Assigned Due Friday, October 23rd
8	24-Sep	Data visualization: data visualization principles, data visualization case study: vaccines	
9	29-Sep	Data visualization: Gapminder case study, time series plots, transformations	
10	1-Oct	Data visualization: Comparing distributions, smooth density plots	
11	6-Oct	Maps: the maps package, plotting global life expectancy, plotting US state murder rates	
12	8-Oct	Maps: COVID-19 case study	
13	13-Oct	Summarizing data: numerical summaries, frequency tables, the **`tableone`** package, reporting missing data	
14	15-Oct	Managing missing data: types of missingness, simple imputation, advanced imputation, survey skip patterns	
15	20-Oct	Hypothesis testing: tests for one group, tests for two groups, tests for more than two groups, nonparametric tests	HW4 Assigned Due Friday, November 6th
16	22-Oct	Introduction to linear regression	
17	27-Oct	Linear regression continued	

18	29-Oct	Introduction to logistic regression	
19	3-Nov	Logistic regression continued	
20	5-Nov	Introduction to machine learning	Midterm assigned November 6th – 15th
21	10-Nov	Machine learning continued	
22	12-Nov	Machine learning continued	
23	17-Nov	Machine learning continued	HW5 Assigned Due Friday, December 4th
24	19-Nov	Machine learning continued	
25	24-Nov	Thanksgiving Recess – No Class	
26	26-Nov	Thanksgiving Recess – No Class	
27	1-Dec	Machine learning continued	
28	3-Dec	Machine learning continued	
29	8-Dec	Machine learning continued	
30	10-Dec	Machine learning continued	
31	15-Dec	Introduction to Shiny	
32	17-Dec	Shiny continued Next steps in data science	Final Project Due Tuesday, December 15th

Note: all lecture topics and assignment dates are subject to change as they rely on the pacing of the course.